

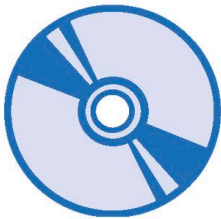
Data storage

The earliest form of data storage for digital computers was on punched card or punched tape. From the 1950s, punched cards were gradually replaced by magnetic tapes and disks. The 1980s saw the introduction of compact discs (CDs), which are read using a reflected laser beam. In the past 20 years, solid-state devices have become the most popular data-storage medium.



△ Punch cards

Data could be entered into early computers using cards with holes punched out of them. In each position on the card a hole represented a 0 and no hole represented a 1.



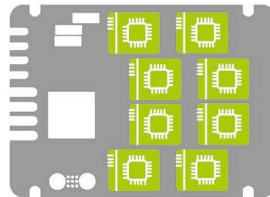
△ Optical disc

Optical discs encode data in tiny indentations on the surface of the disc, called pits and lands. When a laser is shone on the disc, a pit does not reflect the light, and so it represents a 0, while a land reflects the light and represents 1.



△ Magnetic tape

From the 1950s, data was stored on magnetic tape marked in tiny sections, where the opposing polarities represented 0 or 1.



△ Solid-state devices

The latest hard drives are made of solid-state devices (SSDs), chips similar to those that computers use. With no moving parts, they're harder to damage.

REAL WORLD

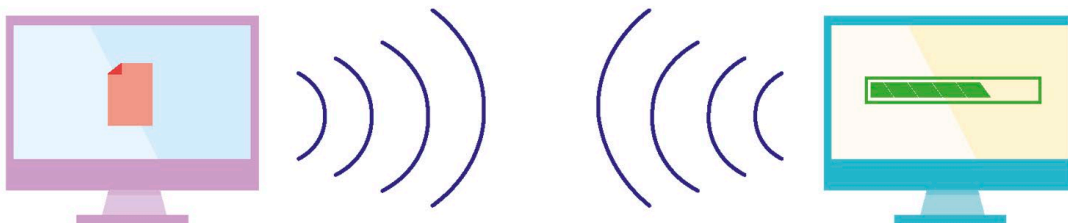
Apollo 11

The data storage of the Apollo 11 rocket used core rope memory, where wires are wound through tiny magnetic rings, called cores, to represent the data. A wire going through the core is read as a 1, while a wire going around it is a 0. Retired female textile workers wove the program by hand. It landed the rocket on the moon. This led to the nickname "little old lady", or "LOL", memory.



Capacity

Hardware capacity is a measure of how much binary code a computer can store. It's usually described by two values: random-access memory (RAM) and hard disk. RAM is the storage used when the computer is executing instructions. The bulk of a computer's data, such as files and programs, are stored on the hard disk. The RAM capacity of a computer is always much smaller than its hard disk capacity.



▽ Connection speeds

This describes how quickly data can be transferred to, or from, a computer. It's measured in megabits per second (Mbps), so 20 Mbps means 20 megabits are transferred each second.